// Day 27 of 30

ARCHITECTURE BREAKDOWN

How Netflix Handles 250M Users

→ Swipe through all 6 slides

Shubham Bhati
linkedin.com/in/bhatishubham

The Scale

Staggering numbers

```
250 million subscribers
190+ countries
17,000+ titles
100M+ hours watched per day

Peak traffic:
  → 400 Gbps of video streaming
  → Thousands of microservices
  → Runs almost entirely on AWS
```

💡 Netflix = largest microservices deployment in the world

Shubham Bhati
linkedin.com/in/bhatishubham

Microservices — 1000+ Services

Each team owns their services

```
API Gateway (Zuul/Spring Cloud Gateway)
  → Routes to correct service

Services:
  → User Profile Service
  → Recommendation Engine
  → Video Encoding Service
  → Billing Service
  → Content Delivery
  → A/B Testing Service

Each: own DB, own team, own deploy
```

💡 Hundreds of deploys per day. No downtime.

Chaos Engineering

Netflix INTENTIONALLY breaks production

```
Chaos Monkey:
  → Randomly kills servers in production
  → Forces teams to build resilient systems

Why?
  → 'If your system can't handle random
     failures, it can't handle real ones'

Results:
  → 99.99% uptime
  → Systems self-heal automatically
```

💡 Netflix invented Chaos Engineering

Shubham Bhati
linkedin.com/in/bhatishubham

Content Delivery — CDN

Video from the nearest server

```
Open Connect CDN:
  → Netflix's own CDN
  → Servers in ISP data centers
  → Video cached at edge
  → User streams from nearest server

Result:
  → Buffer-free streaming
  → Reduced internet traffic by 30%
  → Works even with slow connections
```

💡 Netflix built their OWN CDN. That's commitment.

Recommendation Engine

How it knows what you want

```
80% of what you watch =
  recommended by the algorithm

Inputs:
  → Watch history
  → Time of day
  → Device type
  → What similar users watched
  → How long you watched
  → What you searched

ML models run in real-time
Personalized for each user
```

💡 Recommendations save Netflix $1B/year in retention

Shubham Bhati
linkedin.com/in/bhatishubham

Key Takeaways for Backend Devs

What we can learn

```
1. Build for failure (Chaos Engineering)
2. Microservices enable team autonomy
3. Cache at the edge (CDN)
4. Circuit breakers prevent cascading
5. A/B test everything
6. Observability is non-negotiable
7. Automate deployments completely
```

💡 You don't need Netflix scale to use Netflix patterns

Shubham Bhati
linkedin.com/in/bhatishubham

```
done = true
```

Follow for more!

Like · Comment · Repost

Shubham Bhati
linkedin.com/in/bhatishubham